Calving Fronts summary

Dataset

- 681 samples.
- Locations: Antarctica, Greenland, Alaska (7 glaciers).
- Structure: SAR images, front labels, zones labels, bounding boxes.
- Time span: 1995 – 2020.
- SAR imagery sources: ERS-1/2, Envisat, RADARSAT-1, ALOS Phased Array L-band Synthetic, Aperture Radar (ALOS PALSAR), TerraSAR-X (TSX), TanDEM-X (TDX), and Sentinel-1.
- SAR data sources: German Aerospace Center (DLR), the European Space Agency (ESA), and the Alaska Satellite Facility (ASF).

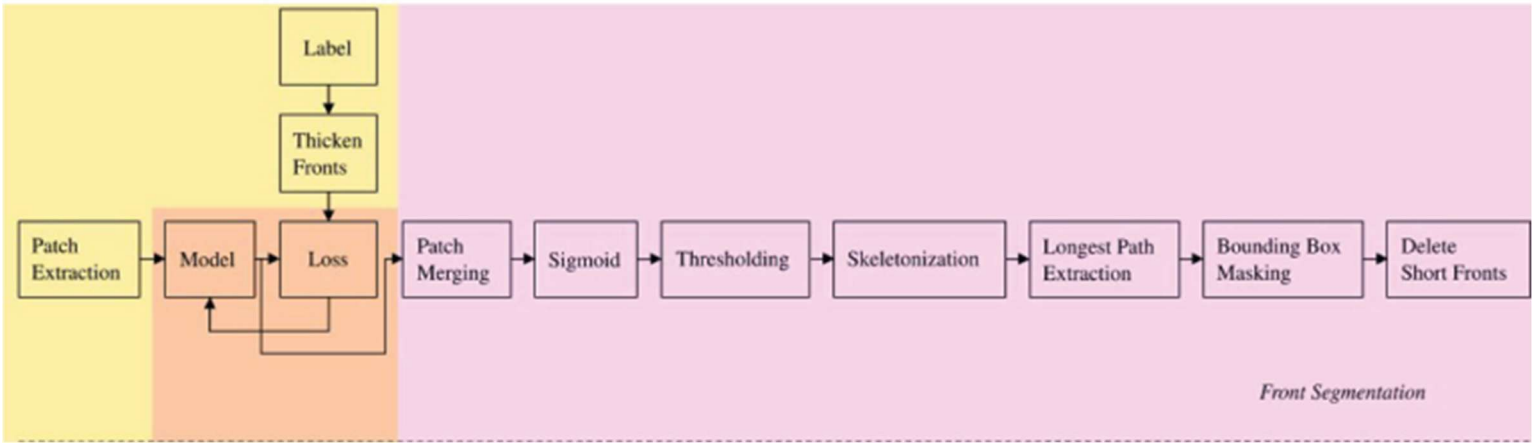
Table 3. Quality factors of manual calving front mapping and the related horizontal uncertainty of the mapped glacier front position.

Quality factor	1	2	3	4	5	6
[m]	70	130	200	230	450	> 450

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Front Segmentation Baseline Method

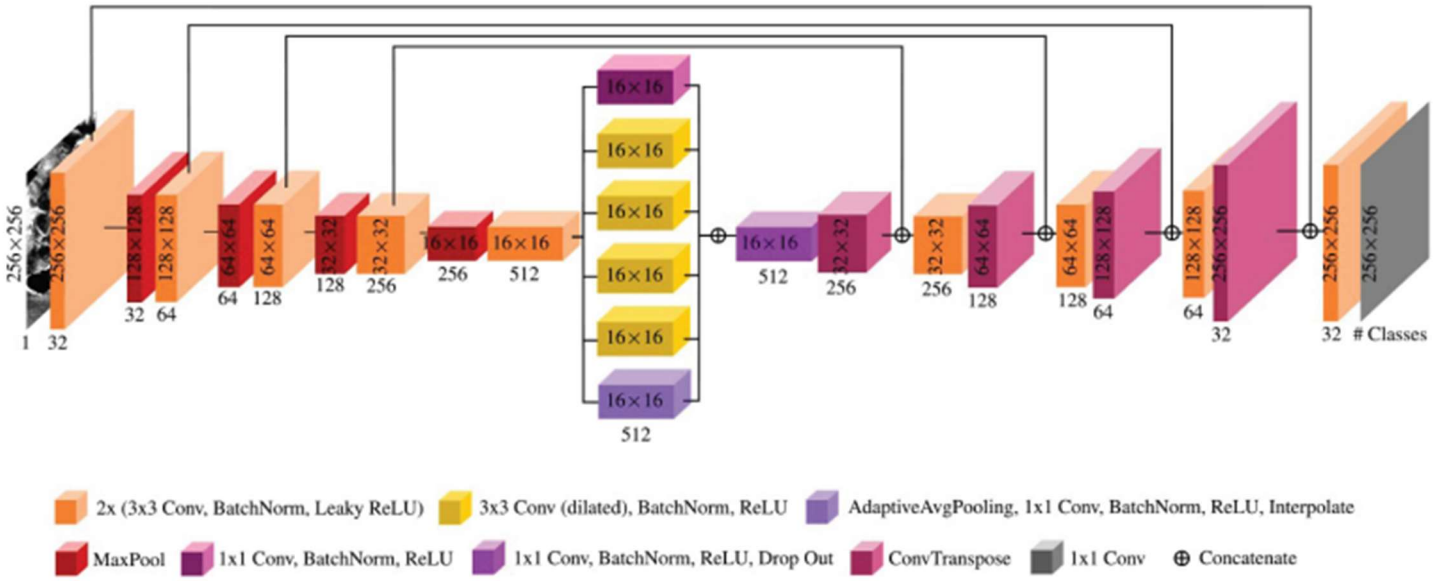
High level scheme:



Preprocessing for front segmentation:

1) Input data and labels gathering.
2) Class balancing with morphological dilation with 5x5 kernel to thicken the fronts and reduce unbalancing issues (since front line is underrepresented with too few pixels).
3) Patch Extraction -> images are divided in 256x256 patches (in training stride with no overlap, in validation and test stride with overlap 128 pixels) and padding with zeros to handle edges.
4) Data augmentation, with fixed probability, with rotations, flips, brightness variation, gaussian noise, wrap (only in training)
5) Normalization with z-score on each patch using mean and standard deviation of training set.

Model:



1) Base architecture:
   - Skip connection Encoder-decoder.
   - Bottleneck enriched with ASPP (Atrous Spatial Pyramid Pooling) equipped with dilation rates 1,2,4,8 (useful since front may have very different scales).
   - Activation: Leaky ReLU (slope 0.1).
   - Padding: “same”.
   - Initial features: 32.
2) Output:
   - Front segmentation: probability map (pixel →probability to be part of the front line)
3) Training setup:
   - Batch size: 16 (max fitting in GPU with 256×256 patch).

- Optimizer: Adam + cyclic learning rate (base/max LR = 1e-4/5e-4 per front).
- Gradient clipping norm = 1.0.
- Front **Loss functions**: adapted Improved Distance Map Loss (Davari 2021) → combining Dice with weights coming from front-line EDT.
- Early stopping with threshold patience 30 epochs.
- Training: 150 epochs × 5 runs → mean of results.

Postprocessing:

- **Patch merging** with Gaussian weighting (more importance on centres, less on edges, predicted overlapped zones are fused via weighted mean and padding is removed).
- **Sigmoid + threshold (0.12)** → binarization.
- **Skeletonization** → 1-pixel lines.
- **Longest path extraction**: deletion of spurious branches.
- **Bounding box masking** + segment filter < 750 m.

Evaluation:

- **Final metric (front delineation)**: **Mean Distance Error (MDE)** → symmetric mean distance between predicted lines and ground truth.

Results with a run of pretrained model

Epochs of pretrained model: 144

Validation:

Average Loss: 0.5756176710128784

Average Precision: 0.026997725059931248

Average Recall: 0.6769877295167577

Average F1 Score: 0.051666733482539196

Average IoU: 0.026549514893114853

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Results for all images

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Number of images: 56

Number of images with no predicted front: 0

Mean distance error: 309.95518257905826

Test:

Average Loss: 0.6058998107910156

Average Precision: 0.029292776000305158

Average Recall: 0.5542650039930039

Average F1 Score: 0.052082547176828135

Average IoU: 0.027040528813207523

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Results for all images

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Number of images: 122

Number of images with no predicted front: 7

Mean distance error: 768.3999955685621